

Note that the calculation for C_{10} only changes in F_e so we can use the following to calculate the new C_{10} value in every iteration:

$$\text{new } C_{10} = \frac{\text{new } F_e}{\text{old } F_e} (\text{old } C_{10})$$

4.4 Shafts and Keys

4.4.1 Design Selection

1. You should be given some shaft with length dimensions and certain gears/sheaves attached as well as bearing locations.

Find the torque from each sheave/gear from the power (P) and angular speed (n)

$$T = \frac{63000P}{n}$$

Verify that the sum of torques is equal to 0

2. Compute the force from each sheave/gear

(a) For spur gears:

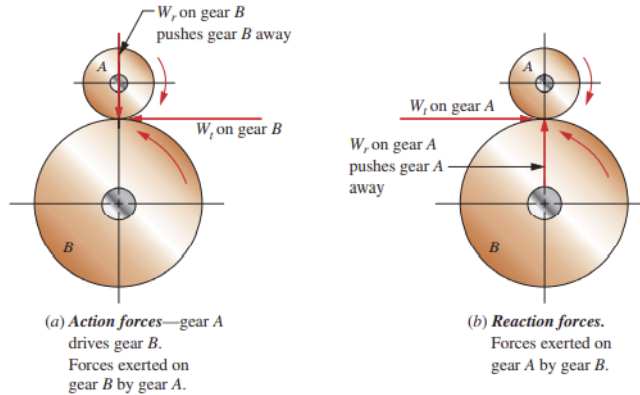
$$\text{tangential force: } W_t = \frac{2T}{D}$$

$$\text{radial force: } W_r = W_t \tan \phi$$

D = pitch diameter

ϕ = pressure angle

The direction will be given by



(b) For helical gears:

$$\text{tangential force: } W_t = \frac{2T}{D}$$

$$\text{radial force: } W_r = W_t \frac{\tan \phi_n}{\cos \psi}$$

$$\text{axial force: } W_x = W_t \tan \psi$$

ϕ_n = normal pressure angle

ψ = helix angle

(c) Bevel gears:

$$\text{tangential force: } W_t = \frac{2T}{D}$$

$$\text{radial load: } W_r = W_t \tan \phi \cos \Gamma = W_t \tan \phi \cos \gamma$$

$$\text{axial load: } W_x = W_t \tan \phi \sin \Gamma = W_t \tan \phi \sin \gamma$$

ϕ = pressure angle

γ = cone angle of pinion

Γ = cone angle of gear

(d) Wormgears:

Refer to wormgear section. I ain't writing that out again

(e) Chain sprockets:

$$F_{\text{shaft}} = \frac{2T}{D}$$

(f) V-Belt sheaves:

$$\frac{F_1}{F_2} = k \text{ (assume } k = 5 \text{ if not given)}$$

$$F_{\text{shaft}} = \frac{2T}{D} \frac{k+1}{k-1}$$

F_1 = tight side tension

F_2 = slack side tension

(g) Flat belt pulleys:

$$\frac{F_1}{F_2} = k \text{ (assume } k = 3 \text{ if not given)}$$

$$F_{\text{shaft}} = \frac{2T}{D} \frac{k+1}{k-1}$$

F_1 = tight side tension

F_2 = slack side tension

3. Take the forces you just calculated and draw a free body diagram of the shaft (recommended to break every force up into x and y components)

4. Calculate the reaction forces at the bearings

Set up the following equations to use:

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum M_x = 0$$

$$\sum M_y = 0$$

$$\sum F_z = 0 \text{ (if axial load)}$$

Note that bearings will only have an axial force component if there's an axial force from a gear.

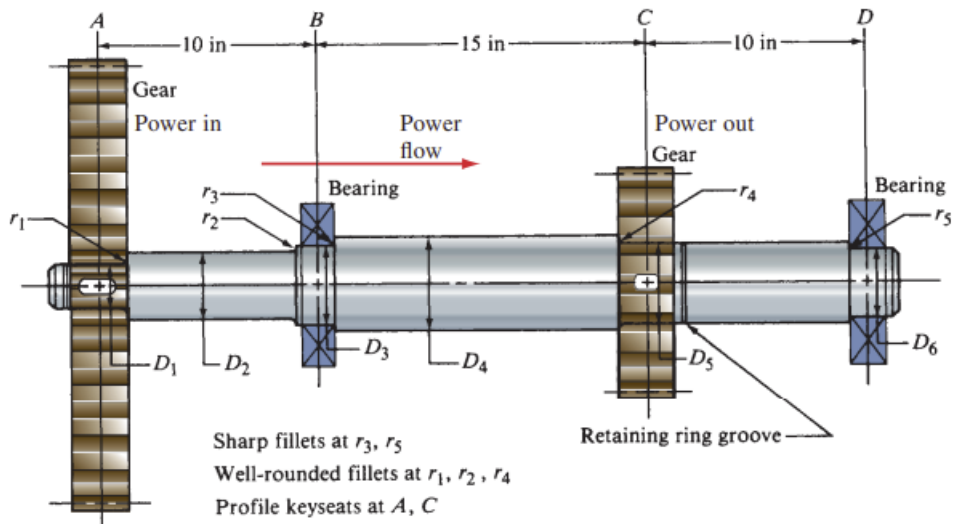
5. Draw the shear diagrams for the horizontal (x) and vertical forces (y)
6. Draw the bending moment diagrams for the horizontal (x) and the vertical (y) moments
Recall bending moment is the integral (area) of the shear force diagram
7. Draw the torque diagram for the shaft (exact same idea as how to draw shear diagram)
Note that we will only have torque gains/drops across gears/pinions and not bearings
8. Compute the resultant shear and moment at each point of interest (where there's a component) along the shaft

$$V = \sqrt{V_x^2 + V_y^2}$$

$$M = \sqrt{M_x^2 + M_y^2}$$

9. Find the stress concentration factors (K_t).

Whenever there's a change in diameter or a component mounted we want to find the K_t value.



We will require K_t values for the following cases (usually corresponds to 3 K_t values for each component accounting for left, center, and right mounting of the component)

- (a) Keyseats:

This corresponds to how the component (gear, pinion, etc.) is fastened in the center (usually a little slot the key slips into to make it turn). This is the center of A and C in the diagram.

For moving parts we have:

- $K_t = 2$ (profile keyseat)

- $K_t = 1.6$ (sled runner keyseat)

We always assume profile unless told otherwise.

(b) Shoulder fillets:

This is how sharp the fillet is when we have a change in diameter.

- $K_t = 2.5$ (sharp fillet)
- $K_t = 1.5$ (well-rounded fillet)

Components such as gears and sheaves can almost always be assumed to have well-rounded fillets. This corresponds to the right of A and the left of C on the diagram.

(c) Fitting bearings:

A bearing will sit on a shoulder and so the diameter of the left, right, and center will all be different.

- Small diameter side:
Use a well-rounded fillet: $K_t = 1.5$
- Bearing seat (middle):
Assume bearing is press fit: $K_t = 1$
- Large diameter side:
Use a sharp fillet: $K_t = 2.5$

(d) Retaining rings:

These will be placed on the small diameter side of components such as gears/pinions to keep them in place.

- $K_t = 3$ (retaining ring)

These are located on the diagram at the left of A and the right of C .

- Specify a material for the shaft and find tensile strength, s_u , and yield strength, s_y .
Tables for material properties can be found at the back of Motts or somewhere in this guide in the gears section.
- Specify a material factor, C_m

Steel type	C_m
Wrought steel:	$C_m = 1.00$
Cast steel:	$C_m = 0.80$
Powdered steel:	$C_m = 0.76$
Malleable cast iron:	$C_m = 0.80$
Gray cast iron:	$C_m = 0.70$
Ductile cast iron:	$C_m = 0.66$

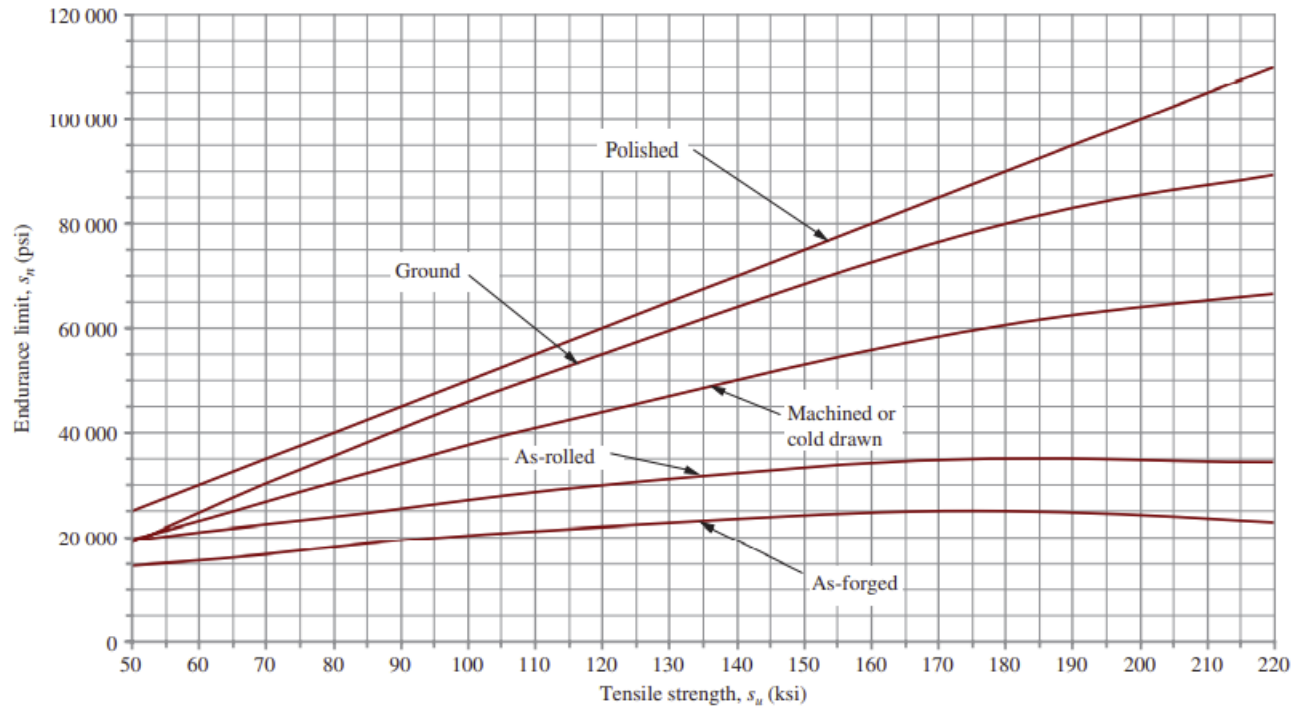
If not specified, assume Wrought steel and $C_m = 1$

- Apply a type-of-stress factor, C_{st}

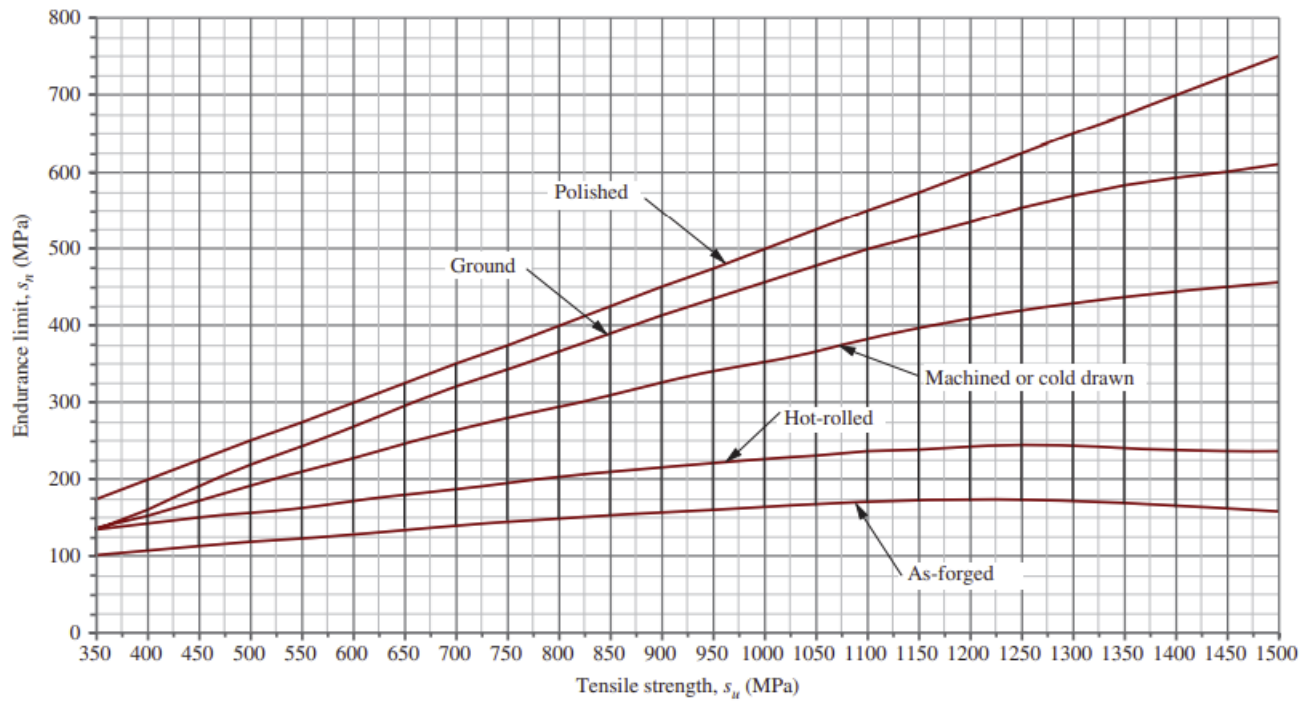
- $C_{st} = 1$ (bending stress)
- $C_{st} = 0.8$ (axial tension)

In almost all cases we will have bending stress so $C_{st} = 1$

13. Based on the material manufacturing (should be given) get s_n from the graph (if not specified assume cold drawn)



(a) U.S. customary units



(b) SI metric units

14. Get the reliability factor C_R , from this table

Desired reliability	C_R
0.50	1.0
0.90	0.90
0.99	0.81
0.999	0.75

A safe guess is 99% reliability so $C_R = 0.81$

15. Guess a size factor C_s .

We will usually want to guess an initial size factor of around $C_s = 0.8$ which corresponds to a shaft diameter of around 2". This will be adjusted later in further iterations if needed

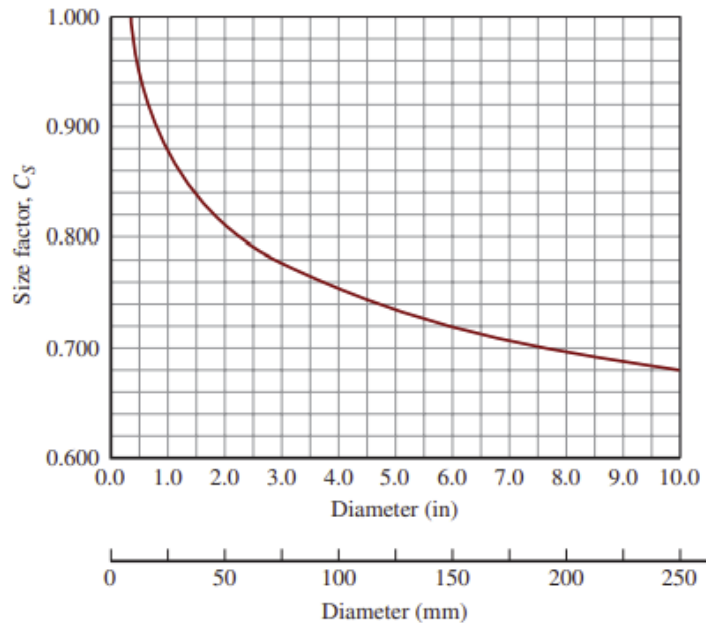


TABLE 5-4 Size Factors	
U.S. customary units	
Size range	For D in inches
$D \leq 0.30$	$C_s = 1.0$
$0.30 < D \leq 2.0$	$C_s = (D/0.3)^{-0.11}$
$2.0 < D < 10.0$	$C_s = 0.859 - 0.021\,25D$
SI units	
Size range	For D in mm
$D \leq 7.62$	$C_s = 1.0$
$7.62 < D \leq 50$	$C_s = (D/7.62)^{-0.11}$
$50 < D < 250$	$C_s = 0.859 - 0.000\,837D$

16. Compute the estimated actual endurance limit s'_n

$$s'_n = s_n C_m C_{st} C_R C_s$$

17. Compute the minimum shaft diameter for each part (i.e. to the left, center, and right of a component. Any place where you may have a different K_t value).
Use this equation if there is torque or bending moment

$$D = \left(\frac{32N}{\pi} \sqrt{\left(\frac{K_t M}{s'_n} \right)^2 + \frac{3}{4} \left(\frac{T}{s_y} \right)^2} \right)^{1/3}$$

Use this equation for if there is no torque or bending (only shear), or if it might exceed the diameter above

$$D = \sqrt{\frac{2.94 K_t V N}{s'_n}}$$

N = design (safety) factor

M = bending moment

T = torque

V = shear force

Take a moment to remember that there are people out there who are proud of you.

18. Multiply any diameters corresponding to retaining rings (with $K_t = 3$) by 1.06

$$D_{\text{retaining ring}} = 1.06D$$

19. Some of these diameters will correspond to the same locations along the shaft. Identify the most restrictive diameters to get a minimum diameter size for each part of the shaft.
20. At this point you may want to consider a 2nd iteration, making a better guess for C_s in step 15. This is recommended if your shaft diameters are significantly different from your initial guess of around 2".

21. Find the C value for the bearing:

- (a) Determine the load life L_2 based on the usage description, basically the total number of revolutions over the expected lifetime:

$$L_2 = (\# \text{ rpm}) \cdot \left(\frac{60\text{min}}{\text{hr}}\right) \left(\frac{\# \text{ hours}}{\text{day}}\right) \left(\frac{\# \text{ days}}{\text{month}}\right) \left(\frac{12\text{month}}{\text{year}}\right) (\# \text{ years})$$

- (b) If the reliability of the bearing is not 90%, then you have to multiply $L_{10} = 10^6$ by a constant to adjust. Find the reliability constant from the table below:

TABLE 14-6 Life Adjustment Factors for Reliability, C_R		
Reliability (%)	C_R	Life designation
90	1.0	L_{10}
95	0.62	L_5
96	0.53	L_4
97	0.44	L_3
98	0.33	L_2
99	0.21	L_1

and your new L_1 value will be:

$$L_1 = C_R \cdot 10^6$$

- (c) Bearing Load/Life relationship, use $k = 3.00$ for ball bearings and $k = 3.33$ for tapered. L_1 is the catalog value of 1×10^6 revs for SKF bearings and most manufacturers. P_2 is the actual radial force experienced on the shaft:

$$\frac{L_2}{L_1} = \left(\frac{C}{P_2}\right)^k$$

$$C = P_2 \left(\frac{L_2}{C_R \cdot 10^6}\right)^{\frac{1}{k}}$$

22. Use the C value and the minimum diameter of the shaft at the bearing to select a suitable bearing from the table

TABLE 14-3 Dimensions for Single Row, Deep-groove Ball Bearings

Bearing number	Nominal bearing dimensions						Basic load ratings				Maximum fillet radius r_{max}^1		Minimum shaft shoulder diameter, S		Maximum housing shoulder diameter, H		Bearing mass	
	Bore, d		Outside dia., D		Width, B		Static, C_0		Dynamic, C									
	mm	in	mm	in	mm	in	kN	lb _f	kN	lb _f	mm	in	mm	in	mm	in	kg	lb _m
6000	10	0.3937	26	1.0236	8	0.3150	1.96	441	4.62	1039	0.3	0.012	12	0.472	24	0.945	0.019	0.042
6200	10	0.3937	30	1.1811	9	0.3543	2.36	531	5.07	1140	0.6	0.024	14	0.551	26	1.024	0.032	0.071
6300	10	0.3937	35	1.3780	11	0.4331	8.06	1812	3.40	764	0.6	0.024	14	0.551	31	1.220	0.053	0.117
6001	12	0.4724	28	1.1024	8	0.3150	2.36	531	5.07	1140	0.3	0.012	14	0.551	26	1.024	0.022	0.049
6201	12	0.4724	32	1.2598	10	0.3937	3.10	697	6.89	1549	0.6	0.024	16	0.630	28	1.102	0.037	0.082
6301	12	0.4724	37	1.4567	12	0.4724	4.15	933	9.75	2192	1.0	0.039	17	0.669	32	1.260	0.060	0.132
6002	15	0.5906	32	1.2598	9	0.3543	2.85	641	5.59	1257	0.3	0.012	17	0.669	30	1.181	0.030	0.066
6202	15	0.5906	35	1.3780	11	0.4331	3.75	843	7.80	1754	0.6	0.024	19	0.748	31	1.220	0.045	0.099
6302	15	0.5906	42	1.6535	13	0.5118	5.40	1214	11.40	2563	1.0	0.039	20	0.787	37	1.457	0.082	0.181
6003	17	0.6693	35	1.3780	10	0.3937	3.25	731	6.05	1360	0.3	0.012	19	0.748	33	1.299	0.039	0.086
6203	17	0.6693	40	1.5748	12	0.4724	4.75	1068	9.56	2149	0.6	0.024	21	0.827	36	1.417	0.065	0.143
6303	17	0.6693	47	1.8504	14	0.5512	6.55	1473	13.50	3035	1.0	0.039	22	0.866	42	1.654	0.120	0.265
6004	20	0.7874	42	1.6535	12	0.4724	5.00	1124	9.36	2104	0.6	0.024	24	0.945	38	1.496	0.069	0.152
6204	20	0.7874	47	1.8504	14	0.5512	6.55	1473	12.70	2855	1.0	0.039	25	0.984	42	1.654	0.110	0.243
6304	20	0.7874	52	2.0472	15	0.5906	7.80	1754	15.90	3575	1.0	0.039	27	1.063	45	1.772	0.140	0.309
6005	25	0.9843	47	1.8504	12	0.4724	6.55	1473	11.20	2518	0.6	0.024	29	1.142	43	1.693	0.080	0.176
6205	25	0.9843	52	2.0472	15	0.5906	7.80	1754	14.00	3147	1.0	0.039	30	1.181	47	1.850	0.130	0.287
6305	25	0.9843	62	2.4409	17	0.6693	11.60	2608	22.50	5058	1.0	0.039	32	1.260	55	2.165	0.230	0.507
6006	30	1.1811	55	2.1654	13	0.5118	8.30	1866	13.30	2990	1.0	0.039	35	1.378	50	1.969	0.160	0.353
6206	30	1.1811	62	2.4409	16	0.6299	11.2	2518	19.5	4384	1.0	0.039	35	1.378	57	2.244	0.200	0.441
6306	30	1.1811	72	2.8346	19	0.7480	16.0	3597	28.1	6317	1.0	0.039	37	1.457	65	2.559	0.350	0.772
6007	35	1.3780	62	2.4409	14	0.5512	10.2	2293	15.9	3575	1.0	0.039	40	1.575	57	2.244	0.160	0.353
6207	35	1.3780	72	2.8346	17	0.6693	15.3	3440	25.5	5733	1.0	0.039	42	1.654	65	2.559	0.290	0.639
6307	35	1.3780	80	3.1496	21	0.8268	19.0	4272	33.2	7464	1.5	0.059	43	1.693	72	2.835	0.460	1.014
6008	40	1.5748	68	2.6772	15	0.5906	11.6	2608	16.8	3777	1.0	0.039	45	1.772	63	2.480	0.190	0.419
6208	40	1.5748	80	3.1496	18	0.7087	19.0	4272	30.7	6902	1.0	0.039	47	1.850	73	2.874	0.370	0.816
6308	40	1.5748	90	3.5433	23	0.9055	24.0	5396	41.0	9218	1.5	0.059	48	1.890	82	3.228	0.630	1.389

TABLE 14-3 Dimensions for Single Row, Deep-groove Ball Bearings (continued)

Bearing number	Nominal bearing dimensions						Basic load ratings				Maximum fillet radius r_{max}^1		Minimum shaft shoulder diameter, S		Maximum housing shoulder diameter, H		Bearing mass	
	Bore, d		Outside dia., D		Width, B		Static, C_0		Dynamic, C									
	mm	in	mm	in	mm	in	kN	lbf	kN	lbf	mm	in	mm	in	mm	in	kg	lb _m
6009	45	1.7717	75	2.9528	16	0.6299	14.6	3282	20.8	4676	1.0	0.039	50	1.969	70	2.756	0.250	0.551
6209	45	1.7717	85	3.3465	19	0.7480	21.6	4856	33.2	7464	1.0	0.039	52	2.047	78	3.071	0.410	0.904
6309	45	1.7717	100	3.9370	25	0.9843	31.5	7082	52.7	11 848	1.5	0.059	53	2.087	92	3.622	0.830	1.830
6010	50	1.9685	80	3.1496	16	0.6299	16.0	3597	21.6	4856	1.0	0.039	55	2.165	75	2.953	0.260	0.573
6210	50	1.9685	90	3.5433	20	0.7874	23.2	5216	35.1	7891	1.0	0.039	57	2.244	83	3.268	0.460	1.014
6310	50	1.9685	110	4.3307	27	1.0630	38.0	8543	61.8	13 894	2.0	0.079	59	2.323	101	3.976	1.050	2.315
6011	55	2.1654	90	3.5433	18	0.7087	21.2	4766	28.1	6317	1.0	0.039	62	2.441	83	3.268	0.390	0.860
6211	55	2.1654	100	3.9370	21	0.8268	29.0	6520	43.6	9802	1.5	0.059	63	2.480	92	3.622	0.610	1.345
6311	55	2.1654	120	4.7244	29	1.1417	45.0	10 117	71.5	16 075	2.0	0.079	64	2.520	111	4.370	1.350	2.977
6012	60	2.3622	95	3.7402	18	0.7087	23.2	5216	29.6	6655	1.0	0.039	67	2.638	88	3.465	0.420	0.926
6212	60	2.3622	110	4.3307	22	0.8661	32.5	7307	47.5	10 679	1.5	0.059	68	2.677	102	4.016	0.780	1.720
6312	60	2.3622	130	5.1181	31	1.2205	52.0	11 691	81.9	18 413	2.0	0.079	71	2.795	119	4.685	1.700	3.749
6013	65	2.5591	100	3.9370	18	0.7087	25.0	5621	30.7	6902	1.0	0.039	72	2.835	93	3.661	0.440	0.970
6213	65	2.5591	120	4.7244	23	0.9055	40.5	9105	55.9	12 567	1.5	0.059	73	2.874	112	4.409	0.990	2.183
6313	65	2.5591	140	5.5118	33	1.2992	60.0	13 489	92.3	20 751	2.0	0.079	76	2.992	129	5.079	2.100	4.631
6014	70	2.7559	110	4.3307	20	0.7874	31.0	6969	37.7	8476	1.0	0.039	77	3.031	103	4.055	0.600	1.323
6214	70	2.7559	125	4.9213	24	0.9449	45.0	10 117	60.5	13 602	1.5	0.059	78	3.071	117	4.606	1.050	2.315
6314	70	2.7559	150	5.9055	35	1.3780	68.0	15 288	104.0	23 381	2.0	0.079	81	3.189	139	5.472	2.500	5.513
6015	75	2.9528	115	4.5276	20	0.7874	33.5	7531	39.7	8925	1.0	0.039	82	3.228	108	4.252	0.640	1.411
6215	75	2.9528	130	5.1181	25	0.9843	49.0	11 016	66.3	14 906	1.5	0.059	83	3.268	122	4.803	1.200	2.646
6315	75	2.9528	160	6.2992	37	1.4567	76.5	17 199	114.0	25 629	2.0	0.079	86	3.386	149	5.866	3.000	6.615
6016	80	3.1496	125	4.9213	22	0.8661	40.0	8993	47.5	10 679	1.0	0.039	87	3.425	118	4.646	0.850	1.874
6216	80	3.1496	140	5.5118	26	1.0236	55.0	12 365	70.2	15 782	2.0	0.079	89	3.504	131	5.157	1.400	3.087
6316	80	3.1496	170	6.6929	39	1.5354	86.5	19 447	124.0	27 878	2.0	0.079	91	3.583	159	6.260	3.600	7.938

TABLE 14-3 Dimensions for Single Row, Deep-groove Ball Bearings (continued)

Bearing number	Nominal bearing dimensions						Basic load ratings				Maximum fillet radius r_{max} ¹		Minimum shaft shoulder diameter, S		Maximum housing shoulder diameter, H		Bearing mass	
	Bore, d		Outside dia., D		Width, B		Static, C_0		Dynamic, C									
	mm	in	mm	in	mm	in	kN	lb _f	kN	lb _f	mm	in	mm	in	mm	in	kg	lb _m
6017	85	3.3465	130	5.1181	22	0.8661	43.0	9667	49.4	11 106	1.0	0.039	92	3.622	123	4.843	0.890	1.962
6217	85	3.3465	150	5.9055	28	1.1024	64.0	14 388	83.2	18 705	2.0	0.079	94	3.701	141	5.551	1.800	3.969
6317	85	3.3465	180	7.0866	41	1.6142	96.5	21 695	133.0	29 901	2.5	0.098	98	3.858	167	6.575	4.250	9.371
6018	90	3.5433	140	5.5118	24	0.9449	50.0	11 241	58.5	13 152	1.5	0.059	98	3.858	132	5.197	1.150	2.536
6218	90	3.5433	160	6.2992	30	1.1811	73.5	16 524	95.6	21 493	2.0	0.079	99	3.898	151	5.945	2.150	4.741
6318	90	3.5433	190	7.4803	43	1.6929	108.0	24 281	143.0	32 149	2.5	0.098	103	4.055	177	6.969	4.900	10.805
6019	95	3.7402	145	5.7087	24	0.9449	54.0	12 140	60.5	13 602	1.5	0.059	103	4.055	137	5.394	1.200	2.646
6219	95	3.7402	170	6.6929	32	1.2598	81.5	18 323	108.0	24 281	2.0	0.079	106	4.173	159	6.260	2.600	5.733
6319	95	3.7402	200	7.8740	45	1.7717	118.0	26 529	153.0	34 397	2.5	0.098	108	4.252	187	7.362	5.650	12.458
6020	100	3.9370	150	5.9055	24	0.9449	54.0	12 140	60.5	13 602	1.5	0.059	108	4.252	142	5.591	1.250	2.756
6220	100	3.9370	180	7.0866	34	1.3386	93.0	20 908	124.0	27 878	2.0	0.079	111	4.370	169	6.654	3.150	6.946
6320	100	3.9370	215	8.4646	47	1.8504	140.0	31 475	174.0	39 119	2.5	0.098	113	4.449	202	7.953	7.000	15.435
6021	105	4.1339	160	6.2992	26	1.0236	65.5	14 726	72.8	16 367	2.0	0.079	114	4.488	151	5.945	1.600	3.528
6221	105	4.1339	190	7.4803	36	1.4173	104.0	23 381	133.0	29 901	2.0	0.079	116	4.567	179	7.047	3.700	8.159
6321	105	4.1339	225	8.8583	49	1.9291	153.0	34 397	182.0	40 917	2.5	0.098	118	4.646	212	8.346	8.250	18.191
6022	110	4.3307	170	6.6929	28	1.1024	73.5	16 524	81.9	18 413	2.0	0.079	119	4.685	161	6.339	1.950	4.300
6222	110	4.3307	200	7.8740	38	1.4961	118.0	26 529	143.0	32 149	2.0	0.079	121	4.764	189	7.441	4.350	9.592
6322	110	4.3307	240	9.4488	50	1.9685	180.0	40 468	203.0	45 638	2.5	0.098	123	4.843	227	8.937	9.550	21.058
6024	120	4.7244	180	7.0866	28	1.1024	80.0	17 986	85.2	19 155	2.0	0.079	129	5.079	171	6.732	2.050	4.520
6224	120	4.7244	215	8.4646	40	1.5748	118.0	26 529	146.0	32 824	2.0	0.079	131	5.157	204	8.031	5.150	11.356
6324	120	4.7244	260	10.2362	55	2.1654	186.0	41 817	208.0	46 763	2.5	0.098	133	5.236	247	9.724	14.500	31.973
6026	130	5.1181	200	7.8740	33	1.2992	100.0	22 482	106.0	23 831	2.0	0.079	139	5.472	191	7.520	3.150	6.946
6226	130	5.1181	230	9.0551	40	1.5748	132.0	29 676	156.0	35 072	2.5	0.098	143	5.630	217	8.543	5.800	12.789
6326	130	5.1181	280	11.0236	58	2.2835	216.0	48 561	229.0	51 484	3.0	0.118	146	5.748	264	10.394	18.000	39.690

¹Maximum fillet on shaft shoulder that will clear radius on bearing race

23. Take the bearing you chose from the table and use its bore diameter to identify the minimum shaft diameter

TABLE 15-5 Shaft and Housing Fits for Bearings
A. Shaft fits

Bearing bore			ISO tolerance grade	Shaft diameter		Limits of fit	
Nominal (mm)	Maximum (in)	Minimum (in)		Maximum (in)	Minimum (in)	Minimum (in)	Maximum (in)
10	0.3937	0.3934	j5	0.3939	0.3936	0.0001L	0.0005T
12	0.4724	0.4721	j5	0.4726	0.4723	0.0001L	0.0005T
15	0.5906	0.5903	j5	0.5908	0.5905	0.0001L	0.0005T
17	0.6693	0.6690	j5	0.6695	0.6692	0.0001L	0.0005T
20	0.7874	0.7870	k5	0.7878	0.7875	0.0001T	0.0008T
25	0.9843	0.9839	k5	0.9847	0.9844	0.0001T	0.0008T
30	1.1811	1.1807	k5	1.1815	1.1812	0.0001T	0.0008T
35	1.3780	1.3775	k5	1.3785	1.3781	0.0001T	0.0010T
40	1.5748	1.5743	k5	1.5753	1.5749	0.0001T	0.0010T
45	1.7717	1.7712	k5	1.7722	1.7718	0.0001T	0.0010T
50	1.9685	1.9680	k5	1.9690	1.9686	0.0001T	0.0010T
55	2.1654	2.1648	k5	2.1660	2.1655	0.0001T	0.0012T
60	2.3622	2.3616	k5	2.3628	2.3623	0.0001T	0.0012T
65	2.5591	2.5585	k5	2.5597	2.5592	0.0001T	0.0012T
70	2.7559	2.7553	k5	2.7565	2.7560	0.0001T	0.0012T
75	2.9528	2.9522	k5	2.9534	2.9529	0.0001T	0.0012T
80	3.1496	3.1490	k5	3.1502	3.1497	0.0001T	0.0012T
85	3.3465	3.3457	k5	3.3472	3.3466	0.0001T	0.0015T
90	3.5433	3.5425	k5	3.5440	3.5434	0.0001T	0.0015T
95	3.7402	3.7394	k5	3.7409	3.7403	0.0001T	0.0015T
100	3.9370	3.9362	k5	3.9377	3.9371	0.0001T	0.0015T
105	4.1339	4.1331	m5	4.1350	4.1344	0.0005T	0.0019T
110	4.3307	4.3299	m5	4.3318	4.3312	0.0005T	0.0019T
115	4.5276	4.5268	m5	4.5287	4.5281	0.0005T	0.0019T
120	4.7244	4.7236	m5	4.7255	4.7249	0.0005T	0.0019T
125	4.9213	4.9203	m5	4.9226	4.9219	0.0006T	0.0023T
130	5.1181	5.1171	m5	5.1194	5.1187	0.0006T	0.0023T
140	5.5118	5.5108	m5	5.5131	5.5124	0.0006T	0.0023T
150	5.9055	5.9045	m6	5.9071	5.9061	0.0006T	0.0026T
160	6.2992	6.2982	m6	6.3008	6.2998	0.0006T	0.0026T
170	6.6929	6.6919	m6	6.6945	6.6935	0.0006T	0.0026T
180	7.0866	7.0856	m6	7.0882	7.0872	0.0006T	0.0026T
190	7.4803	7.4791	m6	7.4821	7.4810	0.0007T	0.0030T
200	7.8740	7.8728	m6	7.8758	7.8747	0.0007T	0.0030T

B. Housing fits

Bearing OD			ISO tolerance grade	Housing bore		Limits of fit	
Nominal (mm)	Maximum (in)	Minimum (in)		Maximum (in)	Minimum (in)	Minimum (in)	Maximum (in)
30	1.1811	1.1807	H8	1.1811	1.1824	0	0.0017L
32	1.2598	1.2594	H8	1.2598	1.2613	0	0.0019L
35	1.3780	1.3776	H8	1.3780	1.3795	0	0.0019L

TABLE 15-5 Shaft and Housing Fits for Bearings (continued)

B. Housing fits							
Bearing OD			ISO tolerance grade	Housing bore		Limits of fit	
Nominal (mm)	Maximum (in)	Minimum (in)		Maximum (in)	Minimum (in)	Minimum (in)	Maximum (in)
37	1.4567	1.4563	H8	1.4567	1.4582	0	0.0019L
40	1.5748	1.5744	H8	1.5748	1.5763	0	0.0019L
42	1.6535	1.6531	H8	1.6535	1.6550	0	0.0019L
47	1.8504	1.8500	H8	1.8504	1.8519	0	0.0019L
52	2.0472	2.0467	H8	2.0472	2.0490	0	0.0023L
62	2.4409	2.4404	H8	2.4409	2.4427	0	0.0023L
72	2.8346	2.8341	H8	2.8346	2.8364	0	0.0023L
80	3.1496	3.1491	H8	3.1496	3.1514	0	0.0023L
85	3.3465	3.3459	H8	3.3465	3.3486	0	0.0027L
90	3.5433	3.5427	H8	3.5433	3.5454	0	0.0027L
100	3.9370	3.9364	H8	3.9370	3.9391	0	0.0027L
110	4.3307	4.3301	H8	4.3307	4.3328	0	0.0027L
120	4.7244	4.7238	H8	4.7244	4.7265	0	0.0027L
125	4.9213	4.9206	H8	4.9213	4.9238	0	0.0032L
130	5.1181	5.1174	H8	5.1181	5.1206	0	0.0032L
140	5.5118	5.5111	H8	5.5118	5.5143	0	0.0032L
150	5.9055	5.9048	H8	5.9055	5.9080	0	0.0032L
160	6.2992	6.2982	H8	6.2992	6.3017	0	0.0035L
170	6.6929	6.6919	H8	6.6929	6.6954	0	0.0035L
180	7.0866	7.0856	H8	7.0866	7.0891	0	0.0035L
190	7.4803	7.4791	H8	7.4803	7.4831	0	0.0040L
200	7.8740	7.8728	H8	7.8740	7.8768	0	0.0040L
215	8.4646	8.4634	H8	8.4646	8.4674	0	0.0040L
225	8.8583	8.8571	H8	8.8583	8.8611	0	0.0040L
230	9.0551	9.0539	H8	9.0551	9.0579	0	0.0040L
240	9.4488	9.4476	H8	9.4488	9.4516	0	0.0040L
250	9.8425	9.8413	H8	9.8425	9.8453	0	0.0040L
260	10.2362	10.2348	H8	10.2362	10.2394	0	0.0046L
270	10.6299	10.6285	H8	10.6299	10.6331	0	0.0046L
280	11.0236	11.0222	H8	11.0236	11.0268	0	0.0046L
290	11.4173	11.4159	H8	11.4173	11.4205	0	0.0046L
300	11.8110	11.8096	H8	11.8110	11.8142	0	0.0046L
310	12.2047	12.2033	H8	12.2047	12.2079	0	0.0046L
320	12.5984	12.5968	H8	12.5984	12.6019	0	0.0051L
340	13.3858	13.3842	H8	13.3858	13.3893	0	0.0051L
360	14.1732	14.1716	H8	14.1732	14.1767	0	0.0051L
380	14.9606	14.9590	H8	14.9606	14.9641	0	0.0051L
400	15.7480	15.7464	H8	15.7480	15.7515	0	0.0051L
420	16.5354	16.5336	H8	16.5354	16.5392	0	0.0056L

NOTE: L = loose; and I = tight.

24. Use the selected bearing shaft diameter to round up any necessary minimum shaft diameters that should be the same or larger than the bearing shaft
25. Round up the shaft diameters (except for the bearing one you chose) to the nearest standard size

TABLE A2-1 Preferred Basic Sizes								
Fractional (in)				Decimal (in)			SI metric (mm)	
1/64	0.015 625	5	5.000	0.010	2.00	8.50	1.0	40
1/32	0.031 25	5 1/4	5.250	0.012	2.20	9.00	1.1	45
1/16	0.0625	5 1/2	5.500	0.016	2.40	9.50	1.2	50
3/32	0.093 75	5 3/4	5.750	0.020	2.60	10.00	1.4	55
1/8	0.1250	6	6.000	0.025	2.80	10.50	1.6	60
5/32	0.156 25	6 1/2	6.500	0.032	3.00	11.00	1.8	70
3/16	0.1875	7	7.000	0.040	3.20	11.50	2.0	80
1/4	0.2500	7 1/2	7.500	0.05	3.40	12.00	2.2	90
5/16	0.3125	8	8.000	0.06	3.60	12.50	2.5	100
3/8	0.3750	8 1/2	8.500	0.08	3.80	13.00	2.8	110
7/16	0.4375	9	9.000	0.10	4.00	13.50	3.0	120
1/2	0.5000	9 1/2	9.500	0.12	4.20	14.00	3.5	140
9/16	0.5625	10	10.000	0.16	4.40	14.50	4.0	160
5/8	0.6250	10 1/2	10.500	0.20	4.60	15.00	4.5	180
11/16	0.6875	11	11.000	0.24	4.80	15.50	5.0	200
3/4	0.7500	11 1/2	11.500	0.30	5.00	16.00	5.5	220
7/8	0.8750	12	12.000	0.40	5.20	16.50	6	250
1	1.000	12 1/2	12.500	0.50	5.40	17.00	7	280
1 1/4	1.250	13	13.000	0.60	5.60	17.50	8	300
1 1/2	1.500	13 1/2	13.500	0.80	5.80	18.00	9	350
1 3/4	1.750	14	14.000	1.00	6.00	18.50	10	400
2	2.000	14 1/2	14.500	1.20	6.50	19.00	11	450
2 1/4	2.250	15	15.000	1.40	7.00	19.50	12	500
2 1/2	2.500	15 1/2	15.500	1.60	7.50	20.00	14	550
2 3/4	2.750	16	16.000	1.80	8.00		16	600
3	3.000	16 1/2	16.500				18	700
3 1/4	3.250	17	17.000				20	800
3 1/2	3.500	17 1/2	17.500				22	900
3 3/4	3.750	18	18.000				25	1000
4	4.000	18 1/2	18.500				28	
4 1/4	4.250	19	19.000				30	
4 1/2	4.500	19 1/2	19.500				35	
4 3/4	4.750	20	20.000					

26. Select a material for the keys and get the yield strength, s_y .
 SAE 1018 Carbon steel is a pretty sexy choice. I'd recommend that one.
 But for the easiest time calculating key length make sure that the key material is weaker than the shaft material.

TABLE 11-4 Examples of Materials Used for Keys

Material designation	Tensile strength s_u		Yield strength s_y	
	(ksi)	(MPa)	(ksi)	(MPa)
Carbon steels (SAE)				
1018	64	441	54	372
1035	72	496	39.5	272
1045	91	627	77	531
1095	140	965	83	572
Alloy steels (SAE)				
4140	102	703	90	621
8630	100	690	95	655
Stainless steels (SAE)				
303	90	621	35	241
304	85	586	35	241
316	85	586	35	241
416	75	517	40	276
Aluminum				
6061	18	124	12	83

Source: Adapted from Internet site 20.

Note: Strength properties typical, not guaranteed.

27. Select a key size using the shaft diameter in the spot of the component

TABLE 11-1 Key Size vs. Shaft Diameter

U.S. inch sizes				SI metric sizes			
Nominal shaft diameter		Key dimensions		Nominal shaft diameter		Key dimensions	
Over (in)	to-including (in)	Width, W (in)	Height, H (in)	Over (mm)	to-including (mm)	Width, W (mm)	Height, H (mm)
0.3125	0.4375	0.09375	0.09375	6	8	2	2
0.4375	0.5625	0.1250	0.1250	8	10	3	3
0.5625	0.875	0.1875	0.1875	10	12	4	4
0.875	1.250	0.2500	0.2500	12	17	5	5
1.250	1.375	0.3125	0.3125	17	22	6	6
1.375	1.75	0.375	0.375	22	30	8	7
1.75	2.25	0.500	0.500	30	38	10	8
2.25	2.75	0.625	0.625	38	44	12	8
2.75	3.25	0.750	0.750	44	50	14	9
3.25	3.75	0.875	0.875	50	58	16	10
3.75	4.50	1.00	1.00	58	65	18	11
4.50	5.50	1.25	1.25	65	75	20	12
5.50	6.50	1.50	1.50	75	85	22	14
6.50	7.50	1.75	1.50	85	95	25	14
7.50	9.00	2.00	1.50	95	110	28	16
9.00	11.00	2.50	1.75	110	130	32	18
11.00	13.00	3.00	2.00	130	150	36	20
13.00	15.00	3.50	2.50	150	170	40	22
15.00	18.00	4.00	3.00	170	200	45	25
18.00	22.00	5.00	3.50	200	230	50	28
22.00	26.00	6.00	4.00	230	260	56	32
26.00	30.00	7.00	5.00	260	290	63	32
				290	330	70	36
				330	380	80	40
				380	440	90	45
				440	500	100	50

Note: Key sizes above the horizontal line are square; others are rectangular.

28. Compute the minimum required length of the key

(a) If the material of the key is weakest and you use a square key then use

$$L_{\min} = \frac{4TN}{DWs_y}$$

T = torque at that point on the shaft

D = shaft diameter

N = safety factor. Assume 3 if not given

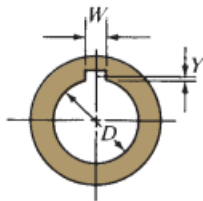
W = the width of the key

(b) If a rectangular key is used or the shaft is the weakest material, use the largest value between the following equations

$$L_{\min} = \frac{4TN}{DHs_y}$$

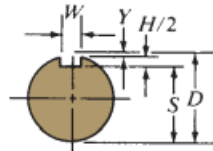
$$L_{\min} = \frac{4TN}{DWs_y}$$

29. Check that your key is a suitable length. Anything larger than 2" may be a little large and you should consider increasing your shaft diameter to compensate. 1-1.5 inches is a good number of inches. The prof suggested under that 1 inch is best in his shaft design video. Or you could say fuck it and not care.
30. Round the key length up to the nearest standard size
31. Compute the measurements for the keyseat and the keyway in the hub



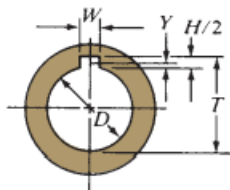
(a) Chordal height

$$Y = \frac{D - \sqrt{D^2 - W^2}}{2}$$



(b) Depth of shaft keyseat

$$S = D - Y - \frac{H}{2} = \frac{D - H + \sqrt{D^2 - W^2}}{2}$$



(c) Depth of hub keyseat

$$T = D - Y + \frac{H}{2} + C = \frac{D + H + \sqrt{D^2 - W^2}}{2} + C$$

Symbols

- C = Allowance
+ 0.005-in clearance for parallel keys
- 0.020-in interference for taper keys
- D = Nominal shaft or bore dia., in
- H = Nominal key height, in
- W = Nominal key width, in
- Y = Chordal height, in